

JIE XUE

PERSONAL DATA

ADDRESS: Office S740, 567 West Yangsi Road,
Pudong District, Shanghai, China 200126
PHONE: +86 19538882526
+1 (612) 8001075
EMAIL: jiexue@nyu.edu
HOMEPAGE: jie-xue.github.io

EMPLOYMENT

AUG 2021 - CURRENT	<i>Assistant Professor of Computer Science</i> New York University Shanghai, China
SEPT 2019 - AUG 2021	<i>Postdoctoral Scholar</i> University of California, Santa Barbara, USA Hosts: Prof. Subhash Suri and Prof. Daniel Lokshantov
JAN 2015 - MAY 2018	<i>Teaching Assistant (part time)</i> University of Minnesota, Twin Cities, USA

EDUCATION

SEPT 2014 - JULY 2019	<i>Doctoral Degree</i> University of Minnesota, Twin Cities, USA Major: Computer Science Minor: Mathematics Research Interests: Computational Geometry Advisor: Prof. Ravi Janardan GPA: 4.0/4.0
SEPT 2011 - APR 2014	<i>Master's Degree</i> Nanjing University of Aeronautics & Astronautics, China Major: Computer Technology Research Interests: Data Mining, Pattern Recognition Advisor: Prof. Songcan Chen GPA: 2.2/4.0
SEPT 2008 - JUNE 2011	<i>Bachelor's Degree</i> Nanjing University of Aeronautics & Astronautics, China Major: Computer Science & Technology GPA: 1.9/4.0

RESEARCH INTERESTS

- Theoretical Computer Science
 - Computational Geometry
 - Algorithms & Data Structures
 - Graph Theory
 - Parameterized Complexity

PUBLICATIONS

(Authors with * are sorted by α - β order. Others are sorted by contribution.)

Surveys and book chapters

1. **Jie Xue***, Meirav Zehavi*, Parameterized algorithms on geometric intersection graphs. To appear in *Computer Science Review*, 2025.

Journal publications

1. Timothy M. Chan*, Qizheng He*, Subhash Suri*, **Jie Xue***, Dynamic geometric set cover, revisited. In *SIAM Journal on Computing*, 2025.
2. Kyungjin Cho*, Eunjin Oh*, Haitao Wang*, **Jie Xue***, Optimal algorithm for the planar two-center problem. In *TheoretCS* (invited from SoCG'24), 2024.
3. Haitao Wang*, **Jie Xue***, Algorithms for halfplane coverage and related problems. In *Discrete & Computational Geometry*, 2024.
4. Sayan Bandyapadhyay*, William Lochet*, Daniel Lokshtanov*, Saket Saurabh*, **Jie Xue***, True contraction decomposition and almost ETH-tight bipartization for unit-disk graphs. In *ACM Transactions on Algorithms*, 2024.
5. Akanksha Agrawal*, Tanmay Inamdar*, Saket Saurabh*, **Jie Xue***, Clustering what matters: optimal approximation for clustering with outliers. In *Journal of Artificial Intelligence Research*, 2023.
6. Pankaj K. Agarwal*, Hsien-Chih Chang*, Subhash Suri*, Allen Xiao*, **Jie Xue***, Dynamic geometric set cover and hitting set. In *ACM Transactions on Algorithms*, 2022.
7. **Jie Xue**, Yuan Li, Rahul Saladi, Ravi Janardan, New bounds for range closest-pair problems. In *Discrete & Computational Geometry*, 2022.
8. Haitao Wang*, **Jie Xue***, Improved algorithms for the bichromatic 2-center problem for pairs of points. In *Computational Geometry: Theory and Applications*, 2021.
9. **Jie Xue**, Yuan Li, Rahul Saladi, Ravi Janardan, Searching for the closest-pair in a query translate. In *Journal of Computational Geometry* (SoCG'19 special issue), 2020.
10. Haitao Wang*, **Jie Xue***, Near-optimal algorithms for shortest paths in weighted unit disk graphs. In *Discrete & Computational Geometry* (SoCG'19 special issue), 2020.
11. Timothy Chan*, Rahul Saladi*, **Jie Xue***, Range closest-pair search in higher dimensions. In *Computational Geometry: Theory and Applications* (WADS'19 special issue), 2020.
12. **Jie Xue**, Yuan Li, Ravi Janardan, Approximate range closest-pair queries. In *Computational Geometry: Theory and Applications* (CCCG'18 special issue), 2020.
13. Yuan Li, Ahmed Eldawy, **Jie Xue**, Nadezda Weber, Mohamed F. Mokbel, Ravi Janardan, Scalable computational geometry in MapReduce. In *VLDB Journal*, 2019.
14. **Jie Xue**, Yuan Li, Ravi Janardan, On the expected diameter, width, and complexity of a stochastic convex-hull. In *Computational Geometry: Theory and Applications*, 2019.
15. Akash Agrawal, Yuan Li, **Jie Xue**, Ravi Janardan, The most-likely skyline problem for stochastic points. In *Computational Geometry: Theory and Applications* (CCCG'17 special issue), 2019.
16. **Jie Xue**, Yuan Li, Ravi Janardan, On the separability of stochastic geometric objects, with applications. In *Computational Geometry: Theory and Applications*, 2018.
17. Yuan Li, **Jie Xue**, Akash Agrawal, Ravi Janardan, On the arrangement of stochastic lines in $\mathbb{R}^2$. In *Journal of Discrete Algorithms*, 2017.

Conference publications

1. Akanksha Agrawal*, Tanmay Inamdar*, **Jie Xue***, Algorithms for simultaneous flip distance between triangulations of polygons. Accepted to the *38th ACM-SIAM Symposium on Discrete Algorithms* (SODA), 2027.

2. Sayan Bandyapadhyay*, William Lochet*, Saket Saurabh*, **Jie Xue***, FPTAS for polygonal multi-robot motion planning. Accepted to the *38th ACM-SIAM Symposium on Discrete Algorithms (SODA)*, 2027.
3. Huairui Chu*, Ajaykrishnan E S*, Daniel Lokshtanov*, Anikait Mundhra*, Thomas Schibler*, Xiaoyang Xu*, **Jie Xue***, Parameterized approximation of rectangle stabbing. In the *34th Annual European Symposium on Algorithms (ESA)*, 2026.
4. Sujoy Bhore*, Subhash Suri*, **Jie Xue***, Xiongxin Yang*, Jiumu Zhu*, Near-optimal dynamic data structures for maximum depth and Klee's measure of boxes. In the *53rd International Colloquium on Automata, Languages, and Programming (ICALP)*, 2026.
5. Sujoy Bhore*, Chih-Hung Liu*, Anurag Murty Naredla*, Yakov Nekrich*, Eunjin Oh*, André van Renssen*, Frank Staals*, Haitao Wang*, **Jie Xue***, Visibility queries in simple polygons. In the *53rd International Colloquium on Automata, Languages, and Programming (ICALP)*, 2026.
6. Thomas Schibler*, **Jie Xue***, Jiumu Zhu*, Approximating convex hulls via range queries. In the *42nd International Symposium on Computational Geometry (SoCG)*, 2026.
7. Daniel Lokshtanov*, Fahad Panolan*, Saket Saurabh*, **Jie Xue***, Meirav Zehavi*, Fine-grained bounds for Courcelle's theorem. In the *58th Annual ACM Symposium on Theory of Computing (STOC)*, 2026.
8. Daniel Lokshtanov*, Saket Saurabh*, **Jie Xue***, The sparsification lemma via measure and conquer. In the *9th SIAM Symposium on Simplicity in Algorithms (SOSA)*, 2026.
9. Thomas Schibler*, Subhash Suri*, **Jie Xue***, Approximation and hardness of polychromatic TSP. In the *37th Canadian Conference on Computational Geometry (CCCG)*, 2025.
10. Sayan Bandyapadhyay*, William Lochet*, Daniel Lokshtanov*, Dániel Marx*, Pranabendu Misra*, Daniel Neuen*, Saket Saurabh*, Prafullkumar Tale*, **Jie Xue***, Robust contraction decomposition for H -minor-free graphs and its applications. In the *52nd International Colloquium on Automata, Languages, and Programming (ICALP)*, 2025.
11. Subhash Suri*, **Jie Xue***, Xiongxin Yang*, Jiumu Zhu*, Dynamic maximum depth of geometric objects. In the *41st International Symposium on Computational Geometry (SoCG)*, 2025.
12. Shinwoo An*, Eunjin Oh*, **Jie Xue***, Single-source shortest path problem in weighted disk graphs. In the *41st International Symposium on Computational Geometry (SoCG)*, 2025. Invited to SoCG special issue (regretfully declined).
13. Thomas Schibler*, Subhash Suri*, **Jie Xue***, Embedding graphs as Euclidean k -NN-graphs. In the *41st International Symposium on Computational Geometry (SoCG)*, 2025.
14. Daniel Lokshtanov*, Fahad Panolan*, Saket Saurabh*, **Jie Xue***, Meirav Zehavi*, Subexponential parameterized algorithms for hitting subgraphs. In the *57th Annual ACM Symposium on Theory of Computing (STOC)*, 2025.
15. Daniel Lokshtanov*, Fahad Panolan*, Saket Saurabh*, **Jie Xue***, Meirav Zehavi*, Efficiently finding and counting patterns with distance constraints in sparse graphs. In the *57th Annual ACM Symposium on Theory of Computing (STOC)*, 2025.
16. Shinwoo An*, Eunjin Oh*, **Jie Xue***, Approximation algorithms for geometric multi-matching problem. In the *57th Annual ACM Symposium on Theory of Computing (STOC)*, 2025.
17. Daniel Lokshtanov*, Fahad Panolan*, Saket Saurabh*, Roohani Sharma*, **Jie Xue***, Meirav Zehavi*, Crossing number in slightly superexponential time. In the *36th ACM-SIAM Symposium on Discrete Algorithms (SODA)*, 2025.
18. Sayan Bandyapadhyay*, Katie Clinch*, William Lochet*, Daniel Lokshtanov*, Saket Saurabh*, **Jie Xue***, PTASes for Euclidean TSP with unit disk and unit square neighborhoods. In the *36th ACM-SIAM Symposium on Discrete Algorithms (SODA)*, 2025.
19. Daniel Lokshtanov*, Abhishek Sahu*, Saket Saurabh*, Vaishali Surianarayanan*, **Jie Xue***, Parameterized approximation for capacitated d -hitting set with hard capacities. In the *36th ACM-SIAM Symposium on Discrete Algorithms (SODA)*, 2025.

20. Viktoriia Korchemna*, Daniel Lokshtanov*, Saket Saurabh*, Vaishali Surianarayanan*, **Jie Xue***, Efficient approximation of hypertree width. In the *65th IEEE Symposium on Foundations of Computer Science (FOCS)*, 2024.
21. Daniel Lokshtanov*, Fahad Panolan*, Saket Saurabh*, **Jie Xue***, Meirav Zehavi*, Bipartizing (pseudo-)disk graphs: approximation with a ratio better than 3. In the *27th International Conference on Approximation Algorithms for Combinatorial Optimization Problems (APPROX)*, 2024.
22. Shinwoo An*, Eunjin Oh*, **Jie Xue***, Sparse outerstring graphs have logarithmic treewidth. In the *32nd Annual European Symposium on Algorithms (ESA)*, 2024.
23. Sayan Bandyapadhyay*, **Jie Xue***, An $O(n \log n)$ -time approximation scheme for Euclidean many-to-many matching. In the *40th International Symposium on Computational Geometry (SoCG)*, 2024. Winner of the Best Paper Award.
24. Haitao Wang*, **Jie Xue***, Algorithms for halfplane coverage and related problems. In the *40th International Symposium on Computational Geometry (SoCG)*, 2024.
25. Kyungjin Cho*, Eunjin Oh*, Haitao Wang*, **Jie Xue***, Optimal algorithm for the planar two-center problem. In the *40th International Symposium on Computational Geometry (SoCG)*, 2024. Invited to TheoretCS.
26. Daniel Lokshtanov*, Fahad Panolan*, Saket Saurabh*, **Jie Xue***, Meirav Zehavi*, A 1.9999-approximation algorithm for vertex cover in string graphs. In the *40th International Symposium on Computational Geometry (SoCG)*, 2024.
27. Timothy M. Chan*, Qizheng He*, **Jie Xue***, Enclosing points with geometric objects. In the *40th International Symposium on Computational Geometry (SoCG)*, 2024.
28. Sayan Bandyapadhyay*, William Lochet*, Daniel Lokshtanov*, Saket Saurabh*, **Jie Xue***, Euclidean bottleneck Steiner tree is fixed-parameter tractable. In the *35th ACM-SIAM Symposium on Discrete Algorithms (SODA)*, 2024.
29. Chinmay Sonar*, Subhash Suri*, **Jie Xue***, Fault tolerance in Euclidean committee selection. In the *31st Annual European Symposium on Algorithms (ESA)*, 2023.
30. Sayan Bandyapadhyay*, William Lochet*, Saket Saurabh*, **Jie Xue***, Minimum-membership geometric set cover, revisited. In the *39th International Symposium on Computational Geometry (SoCG)*, 2023.
31. Akanksha Agrawal*, Tanmay Inamdar*, Saket Saurabh*, **Jie Xue***, Clustering what matters: optimal approximation for clustering with outliers. In the *37th AAAI conference on Artificial Intelligence (AAAI)*, 2023. Selected as AAAI Distinguished Paper.
32. Daniel Lokshtanov*, Fahad Panolan*, Saket Saurabh*, **Jie Xue***, Meirav Zehavi*, A framework for approximation schemes on disk graphs. In the *34th ACM-SIAM Symposium on Discrete Algorithms (SODA)*, 2023.
33. Rong Gu, Han Li, Haipeng Dai, Wenjie Huang, **Jie Xue**, Meng Li, Jiaqi Zheng, Haoran Cai, Yihua Huang, Guihai Chen, ShadowAQP: efficient approximate group-by and join query via attribute-oriented sample size allocation and data generation. In the *49th International Conference on Very Large Data Bases (VLDB)*, 2023.
34. Chinmay Sonar*, Subhash Suri*, **Jie Xue***, Multiwinner elections under minimax Chamberlin-Courant rule in Euclidean space. In the *31st International Joint Conference on Artificial Intelligence (IJCAI)*, 2022.
35. Sayan Bandyapadhyay*, William Lochet*, Daniel Lokshtanov*, Saket Saurabh*, **Jie Xue***, True contraction decomposition and almost ETH-tight bipartization for unit-disk graphs. In the *38th International Symposium on Computational Geometry (SoCG)*, 2022.
36. Neeraj Kumar*, Daniel Lokshtanov*, Saket Saurabh*, Subhash Suri*, **Jie Xue***, Point separation and obstacle removal by finding and hitting odd cycles. In the *38th International Symposium on Computational Geometry (SoCG)*, 2022. Invited to SoCG special issue (regretfully declined).

37. Sayan Bandyapadhyay*, William Lochet*, Daniel Lokshtanov*, Saket Saurabh*, **Jie Xue***, Subexponential parameterized algorithms for cut and cycle hitting problems on H-minor-free graphs. In the *33rd ACM-SIAM Symposium on Discrete Algorithms (SODA)*, 2022.
38. Timothy M. Chan*, Qizheng He*, Subhash Suri*, **Jie Xue***, Dynamic geometric set cover, revisited. In the *33rd ACM-SIAM Symposium on Discrete Algorithms (SODA)*, 2022.
39. Daniel Lokshtanov*, Fahad Panolan*, Saket Saurabh*, **Jie Xue***, Meirav Zehavi*, Subexponential parameterized algorithms on disk graphs. In the *33rd ACM-SIAM Symposium on Discrete Algorithms (SODA)*, 2022.
40. Daniel Lokshtanov*, Saket Saurabh*, Subhash Suri*, **Jie Xue***, An ETH-tight algorithm for multi-team formation. In the *41st Conference on Foundations of Software Technology and Theoretical Computer Science (FSTTCS)*, 2021.
41. Daniel Lokshtanov*, Subhash Suri*, **Jie Xue***, Efficient algorithms for least square piecewise polynomial regression. In the *29th Annual European Symposium on Algorithms (ESA)*, 2021.
42. Zhenyu Pan, **Jie Xue**, Tingjian Ge, Intuitive searching: an approach to search the decision policy of a Blackjack agent. In the *6th International Congress on Information and Communication Technology (ICICT)*, 2021.
43. Daniel Lokshtanov*, Chinmay Sonar*, Subhash Suri*, **Jie Xue***, Fair covering of points by balls. In the *32nd Canadian Conference on Computational Geometry (CCCG)*, 2020.
44. Pankaj K. Agarwal*, Hsien-Chih Chang*, Subhash Suri*, Allen Xiao*, **Jie Xue***, Dynamic geometric set cover and hitting set. In the *36th International Symposium on Computational Geometry (SoCG)*, 2020. Invited to SoCG special issue (regretfully declined).
45. Haitao Wang*, **Jie Xue***, Improved algorithms for the bichromatic 2-center problem for pairs of points. In the *16th Algorithms and Data Structures Symposium (WADS)*, 2019.
46. Timothy Chan*, Rahul Saladi*, **Jie Xue***, Range closest-pair search in higher dimensions. In the *16th Algorithms and Data Structures Symposium (WADS)*, 2019. Invited to WADS special issue.
47. **Jie Xue**, Yuan Li, Rahul Saladi, Ravi Janardan, Searching for the closest-pair in a query translate. In the *35th International Symposium on Computational Geometry (SoCG)*, 2019. Invited to SoCG special issue.
48. Haitao Wang*, **Jie Xue***, Near-optimal algorithms for shortest paths in weighted unit disk graphs. In the *35th International Symposium on Computational Geometry (SoCG)*, 2019. Invited to SoCG special issue.
49. **Jie Xue**, Colored range closest-pair problem under general distance functions. In the *30th ACM-SIAM Symposium on Discrete Algorithms (SODA)*, 2019.
50. **Jie Xue**, Yuan Li, Ravi Janardan, Approximate range closest-pair queries. In the *30th Canadian Conference on Computational Geometry (CCCG)*, 2018. Invited to CCCG special issue.
51. **Jie Xue**, Yuan Li, Rahul Saladi, Ravi Janardan, New bounds for range closest-pair problems. In the *34th International Symposium on Computational Geometry (SoCG)*, 2018.
52. Zhenyu Pan, **Jie Xue**, Yang Gao, Honghao Wang, Guanling Chen, Revealing the relations between learning behaviors and examination scores via a prediction system. In the *2nd International Conference on Computer Science and Artificial Intelligence (CSAI)*, 2018.
53. **Jie Xue**, Yuan Li, Ravi Janardan, On the expected diameter, width, and complexity of a stochastic convex-hull. In the *15th Algorithms and Data Structures Symposium (WADS)*, 2017.
54. **Jie Xue**, Yuan Li, Stochastic closest-pair problem and most-likely nearest-neighbor search in tree spaces. In the *15th Algorithms and Data Structures Symposium (WADS)*, 2017.

55. Akash Agrawal, Yuan Li, **Jie Xue**, Ravi Janardan, The most-likely skyline problem for stochastic points. In the *29th Canadian Conference on Computational Geometry (CCCG)*, 2017. Invited to CCCG special issue.
56. **Jie Xue**, Yuan Li, Ravi Janardan, On the separability of stochastic geometric objects, with applications. In the *32nd International Symposium on Computational Geometry (SoCG)*, 2016.

Manuscripts

1. Madhumita Kundu*, Daniel Lokshtanov*, Saket Saurabh*, **Jie Xue***, Capacitated set cover, hitting set, and domination in the semiladder-free setting. In progress.
2. Zdeněk Dvořák*, Daniel Lokshtanov*, Fahad Panolan*, Saket Saurabh*, **Jie Xue***, Meirav Zehavi*, Efficient approximation for subgraph-hitting problems in sparse graphs and geometric intersection graphs. In progress.
3. Sayan Bandyapadhyay*, William Lochet*, Daniel Lokshtanov*, Saket Saurabh*, Chinmay Sonar*, Vaishali Surianarayanan*, **Jie Xue***, How to fix a broken metric: a linear kernel, tight bounds, and tree metrics. In progress.
4. Daniel Lokshtanov*, Fahad Panolan*, Saket Saurabh*, **Jie Xue***, Meirav Zehavi*, Fine-grained MSO model checking for cliquewidth. In progress.
5. Yuxi Liu*, Saket Saurabh*, Kangyi Tian*, Mingyu Xiao*, **Jie Xue***, Constant-factor FPT approximation for rectangle hitting set and rectangle set cover. In progress.
6. Sayan Bandyapadhyay*, William Lochet*, Saket Saurabh*, **Jie Xue***, Euclidean capacitated sum-of-radii clustering: hardness and approximation. In progress.
7. Huairui Chu*, Ajaykrishnan E S*, Daniel Lokshtanov*, Viktoriia Korchemna*, Saket Saurabh*, Thomas Schibler*, Xiaoyang Xu*, **Jie Xue***, Xiongxin Yang*, Exponential time approximation schemes for layout problems. In progress.
8. Ajaykrishnan E S*, Daniel Lokshtanov*, Saket Saurabh*, **Jie Xue***, Algorithms for min-color path with bounded color-degree. In progress.
9. Sujoy Bhore*, **Jie Xue***, Xiongxin Yang*, Jiumu Zhu*, Simple and efficient algorithms for approximating maximum depth of unit disks and unit balls. In progress.
10. Sujoy Bhore*, **Jie Xue***, Xiongxin Yang*, Jiumu Zhu*, Approximate diameter in dynamic unit disk graphs. In progress.
11. **Jie Xue**, Yuan Li, On dominance-free samples of a (colored) stochastic dataset. In progress.

TALKS

Invited talks

- Talk at Chengdu A&L Seminar (June 2026):
Fine-grained bounds for Courcelle's theorem.
- Talk at NYU Shanghai (April 2026):
Efficient algorithms for geometric and graph problems.
- Tutorial talk at MTPC Workshop (April 2026):
Parameterized problems and algorithms in computational geometry.
- Talk at MTPC Workshop 2026 (April 2026):
Fine-grained bounds for Courcelle's theorem.
- Talk at UCSB (Jan. 2026):
Dynamic maximum depth and Klee's measure.
- Talk at NYU Theory Seminar (May 2025):
Subexponential parameterized algorithms for hitting subgraph.
- Talk at Courant Geometry Seminar (April 2025):
Efficient approximation algorithms for geometric many-to-many matching.

- Talk at Shanghai Theory Day (March 2025):
Subexponential parameterized algorithms for hitting subgraph.
- Talk at Lorentz Workshop (Feb. 2025):
Efficient approximation algorithms for geometric many-to-many matching.
- Talk at Dagstuhl 24411 (Oct. 2024):
New developments in parameterized computational geometry.
- Talk at IJTCS-FAW 2024:
A 1.9999-approximation algorithm for vertex cover in string graphs.
- Talk at CCF National Conference of TCS 2024:
An $O(n \log n)$ -time approximation scheme for Euclidean many-to-many matching.
- Talk at UCSB (Oct. 2023):
Vertex deletion on disk graphs.
- Talk at Nanjing University (Aug. 2022):
Vertex deletion on disk graphs.
- Talk at NYU Tandon (Online, Nov. 2021):
Efficient algorithms and data structures for geometric computing.
- Talk at NYU Shanghai (Online, March 2021):
Efficient algorithms and data structures for geometric computing.
- Talk at Nanjing University (Online, Dec. 2020):
Efficient algorithms and data structures for geometric computing.
- Talk at Nanjing University (Aug. 2018):
Range closest-pair search.

Conference presentations

- Presentation at SoCG 2024 (Online):
 1. An $O(n \log n)$ -time approximation scheme for Euclidean many-to-many matching.
 2. A 1.9999-approximation algorithm for vertex cover in string graphs.
- Presentation at SoCG 2023:
Minimum-membership geometric set cover, revisited.
- Presentation at SODA 2023 (Online):
A framework for approximation schemes on disk graphs.
- Presentation at SoCG 2022 (Online):
Point separation and obstacle removal by finding and hitting odd cycles.
- Presentation at SODA 2022 (Online):
 1. Subexponential parameterized algorithms for cut and cycle hitting problems on H-minor-free graphs.
 2. Subexponential parameterized algorithms on disk graphs.
- Presentation at ESA 2021 (Online):
Efficient algorithms for least square piecewise polynomial regression.
- Presentation at SoCG 2020:
Dynamic geometric set cover and hitting set.
- Presentation at SoCG 2019:
 1. Searching for the closest-pair in a query translate.
 2. Near-optimal algorithms for shortest paths in weighted unit disk graphs.
- Presentation at SODA 2019 and FWCG 2018:
Colored range closest-pair problem under general distance functions.
- Presentation at CCCG 2018:
Approximate closest-pair search.
- Presentation at SoCG 2018:
New bounds for range closest-pair problems.

- Presentation at Young Researcher Forum in CG Week 2018:
Searching for the closest-pair in a convex polygonal translate.
- Presentation at WADS 2017:
 1. Stochastic closest-pair problem and most-likely NN search in tree spaces.
 2. On the expected diameter, width, and complexity of a stochastic convex-hull.
- Presentation at SoCG 2016:
On the separability of stochastic geometric objects, with applications.

STUDENTS

PhD students

- Jiumu Zhu (2024-now)

Undergraduate students advised

- Xiongxin Yang (2024-2025)
- Junqiu Hu (2025-present)

PROFESSIONAL ACTIVITIES AND SERVICES

Workshop organizer

- Parameterized Algorithms for Geometric Problems (at SoCG 2023)

Editorial board of journals

- Journal of Computational Geometry (2026-Current)
- DCG special issue of SoCG'25 (2025)

Program committee of conferences

- ICALP 2027
- ITCS 2027
- SOSA 2027
- CCCG 2026
- SoCG 2025
- ESA (Track A) 2025
- WADS 2023, 2025
- IPEC 2025
- STACS 2024, 2027
- FAW 2022
- CG Week YRF 2021

Reviewer for journals

- Journal of the ACM
- SIAM Journal on Computing
- Discrete & Computational Geometry
- Journal of Computational Geometry
- Algorithmica
- Computational Geometry: Theory and Applications
- Theoretical Computer Science
- Computing in Geometry and Topology

- Journal of Combinatorial Optimization
- International Journal of Computational Geometry and Applications
- VLDB Journal

Reviewer for conferences

- STOC 2023, 2026
- FOCS 2021, 2024, 2025, 2026
- SODA 2021, 2022, 2023, 2024, 2025, 2026, 2027
- SoCG 2017, 2020, 2022, 2023, 2024, 2026
- ICALP 2021, 2022, 2024, 2025, 2026
- ESA 2022, 2023, 2026
- APPROX 2023, 2026
- ISAAC 2018, 2020, 2021, 2023, 2024, 2026
- WADS 2017
- SWAT 2020
- FSTTCS 2021, 2024
- MFCS 2017
- IPEC 2023, 2026
- FAW 2019
- COCOON 2023
- COCOA 2021

Services at NYU Shanghai

- CSDSE curriculum committee 2026-2027
- CSDSE faculty search committee 2025-2026
- PC for Capstone projects 2021, 2022, 2023
- Mentor for DURF projects 2022, 2023, 2026

AWARDS

- SoCG Best Paper Award (2024)
- AAAI Distinguished Paper Award (2023)
- Shanghai Leading Talents Program (2023)
- SoCG Travel Award (2016, 2018, 2019)
- SIAM Travel Award (2018)
- Doctoral Dissertation Fellowship at University of Minnesota (2018-2019)

TEACHING

At NYU Shanghai:

- CSCI-SHU 2314 - Discrete Math (Fall 2024, Fall 2025, Spring 2026, Fall 2026)
- CSCI-SHU 220 - Algorithms (Spring 2022, Spring 2023, Spring 2024)
- CSCI-SHU 210 - Data Structures (Spring 2023)
- CSCI-SHU 11 - Introduction to Computer Programming (Fall 2021, Fall 2022)

At University of Minnesota (as Teaching Assistant):

- CSci 5421 - Advanced Algorithms and Data Structures (Spring 2015, Fall 2017, Spring 2018)
- CSci 4011 - Formal Languages and Automata Theory (Spring 2016, Fall 2016, Spring 2017)
- CSci 2011 - Discrete Structures of Computer Science (Fall 2015)

LANGUAGES

- Chinese (traditional preferred)
- English (US)

HOBBIES

- Calligraphy
- Poetry (traditional)
- Reading
- Table-tennis
- Video games (RPG, RTS, MOBA, etc.)